

Waste Whisper

Confidence-Aware Multimodal Waste Classification with IoT Monitoring and Human-in-the-Loop Learning

1. One-Line Pitch

Waste Whisper is a smart mini conveyor that doesn't just guess what trash is — it **knows when it doesn't know**. When the camera is unsure, it "asks" physical sensors (weight, metal-detection, material response) for a second opinion before sorting, and logs everything it's unsure about so a human can teach it later.

This "confidence-aware" framing is your strongest hook for judges — lead with it in your title slide and abstract.

2. Objectives

- Build a functional mini conveyor with automatic servo-based diversion into 4 outputs (Paper / Plastic / Metal / Manual).
 - Train a CV model to classify waste with a calibrated **confidence score**, not just a raw label.
 - Build a **decision engine**: high confidence → auto-sort; medium confidence → pull in sensor evidence; low confidence → reject to manual bin.
 - Fuse camera output with inductive, capacitive, and load-cell sensor readings for ambiguous items.
 - Build a live IoT dashboard: throughput, class counts, confidence trends, bin fill levels.
 - Implement a human-in-the-loop labelling flow so rejected/uncertain items feed back into the training set.
 - Run a controlled experiment comparing **vision-only vs. sensor-only vs. multimodal fusion** on accuracy, wrong-sort rate, and auto-accept rate — this is your "research contribution" section.
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3. System Architecture

Object → Entry Sensor (IR) → Conveyor Belt → Camera Capture

→ Vision AI (class + confidence)

→ Confidence Gate:

HIGH → Direct Sort

MEDIUM → Trigger Sensor Fusion (inductive + capacitive + load cell)

LOW → Route to Manual/Uncertain Bin

- Final Decision → ESP32 → Servo Diverter Gate
- Bin (Paper / Plastic / Metal / Manual)
- Logging: prediction, confidence, sensor readings, final class, timestamp
- SQLite DB → Backend API → Live Dashboard
- Manual Bin items → Human labelling → Retraining dataset

4. Hardware & Software Stack

Component	Qty	Purpose
ESP32 DevKit	1	Sensor reading, motor/servo control, Wi-Fi/IoT link
USB Webcam	1	Image capture for classification
DC geared motor + driver	1	Belt movement
High-torque servos	3	Diverter gates (one per junction)
Inductive sensor	1	Metal detection
Capacitive sensor	1	Material response (helps separate plastic/paper)
Load cell + HX711	1	Weight measurement
IR/photoelectric sensors	2–3	Object entry + position tracking
Ultrasonic sensors	3	Bin fill-level monitoring

Conveyor frame, belt, rollers 1 set Mechanical base

Software: Python, OpenCV, PyTorch or TensorFlow, Arduino IDE/PlatformIO, FastAPI or Flask, SQLite, React or Streamlit for the dashboard. Serial for prototype-stage comms; MQTT/HTTP over Wi-Fi once IoT layer is live.

5. Budget Estimate

Build tier	Cost (₹)	Notes
Functional prototype (laptop-assisted)	6,500 – 9,000	Good enough for internal demo/testing
Presentation-quality build	9,000 – 12,000	Cleaner frame, better wiring, demo-day ready

Assumes an existing laptop handles inference, backend, and dashboard — no dedicated edge device needed initially.

6. Team Structure & Task Allocation (4 Members)

Rather than rigid silos, structure it as **owned domains with shared integration checkpoints** — this avoids the classic "everyone builds their piece, nothing connects" failure mode in 4-person hardware projects.

Member 1 — AI/ML Lead

Owens: the "brain" of the system.

- Collect/curate a waste image dataset (Paper / Plastic / Metal), or fine-tune on an existing open dataset (e.g., TrashNet-style) plus your own captured images.
- Train the classifier (start with a lightweight model — MobileNetV2 or ResNet18 transfer learning — for laptop-speed inference).
- Implement **confidence calibration** (don't just use raw softmax — consider temperature scaling so "confidence" is actually meaningful).
- Define the confidence thresholds (high/medium/low) in collaboration with Member 4's experiments.
- Build the sensor-fusion decision logic (start rule-based/weighted, upgrade to a small classifier later if time permits).
- Own the active-learning/retraining pipeline for manual-bin items.

Member 2 — Embedded/IoT Lead

Owens: everything that touches the ESP32.

- Wire and code IR entry sensors, inductive/capacitive sensors, load cell (HX711), ultrasonic bin sensors.
- Write ESP32 firmware for motor control, servo gate timing, and object travel-time tracking.
- Implement the Serial (prototype) and later MQTT/HTTP (Wi-Fi IoT) communication layer to the laptop/backend.
- Handle sensor calibration — this is the most fiddly part of the whole build, budget real time for it.
- Coordinate closely with Member 1 on the data format sensors send (so fusion logic gets clean numbers).

Member 3 — Mechanical/Electronics Lead

Owens: the physical build.

- Design and build the conveyor frame, belt, rollers, and gate mechanisms.
- Mount servos, sensors, and camera at correct positions/angles for reliable capture and diversion timing.
- Handle power distribution and wiring (motor driver, servo power, sensor power — keep noisy motor lines separate from sensor lines).
- Work with Member 2 on physical placement vs. sensor read reliability (e.g., inductive sensor standoff distance).

- Own the "does it actually work mechanically 50 times in a row" reliability testing.

Member 4 — Backend/Research Lead

Owens: data, dashboard, and the story you tell judges.

- Build the SQLite schema (predictions, confidence, sensor readings, final class, timestamps, bin levels).
- Build the FastAPI/Flask backend that logs from ESP32/laptop and serves the dashboard.
- Build the live dashboard (React or Streamlit): throughput, class distribution, confidence histogram, bin-fill gauges.
- Design and run the **vision-only vs. sensor-only vs. multimodal** comparison experiment — this is your differentiator, don't leave it to the last week.
- Own the final report/documentation and presentation narrative.

Shared checkpoints (all 4): integration days after each phase in the plan below — don't let subsystems stay isolated until week 10.

7. Development Plan (Suggested Phasing)

Phase Focus		Primary Owner(s)
1	Conveyor mechanics built and belt runs reliably	Member 3
2	Camera-only classifier working on laptop	Member 1
3	ESP32 + servo gates respond to a manual trigger	Member 2
4	Physical sensors wired and calibrated	Member 2 + 3
5	Confidence gating + sensor fusion logic integrated	Member 1 + 2
6	IoT dashboard live, logging real runs	Member 4
7	Manual labelling / active-learning loop added	Member 1 + 4
8	Controlled experiments (3-way comparison) run	Member 4
9	Reliability testing, edge-case handling, polish	All
10	Final report + demo rehearsal	All

Run a full integration test at the end of **every** phase, not just at the end — hardware projects die from late integration.

8. Research/Innovation Angle (Your Strongest Card)

Most student waste-sorters are "camera → CNN → servo." Yours adds a genuinely defensible research question:

When should an autonomous system trust its own vision, when should it seek more evidence, and when should it defer to a human?

Evaluate the same test set three ways and report:

- **Accuracy, Precision, Recall, F1** per configuration
- **Wrong-sort rate** (the metric that actually matters for real recycling — a confident wrong answer is worse than a rejected one)
- **Auto-accept rate** (% of items the system handled without needing sensors or a human)

This turns "we built a sorter" into "we quantified the cost/benefit of multimodal confidence-gating in waste sorting" — much stronger for a final-year evaluation.

9. Scope & Limitations (Be Upfront About These)

- First prototype handles separated, lightweight, single-material objects — not mixed/composite waste.
 - Camera accuracy depends on lighting, angle, and dataset quality.
 - Inductive/capacitive sensors need per-material calibration and won't be perfect.
 - Multilayer packaging (e.g., chip packets) will likely stay in the "uncertain" bucket — that's fine, call it out as expected behavior, not a bug.
 - Treat advanced active learning and richer fusion models as stretch goals after the core loop works end-to-end.
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10. Expected Deliverables

1. Working mini conveyor sorting Paper/Plastic/Metal + Manual bin
2. Confidence-aware multimodal decision system
3. Live IoT dashboard
4. Experimental comparison report (vision-only vs. sensor-only vs. fusion)
5. Labelled dataset of rejected/uncertain items
6. Final documentation + demo